Mathematics practice booklet: Grade 4 expectations

Grade 4 · Practice booklet

Name: _____

What this booklet is

This is general practice written for Mathematics at Grade 4 level, following the Common Core State Standards for Mathematics standards for that grade. It is not made from your child's MAP questions, and it does not replace the classroom. Use it as a guide to what a child is expected to know and do at this level.

How to use it

- Do one section at a time, about 10 minutes, 3 days a week. Stop while it is still enjoyable.
- Read each question all the way through, twice, before answering. Underline the key words and say what is being asked.
- Take your time. A careful, correct answer is worth more than a fast one.
- Use scrap paper and show the working. Then check the answer before moving on.
- Ask your child to explain how they got an answer, even when it is right.
- If a section feels too hard, do the section before it again, or ask the teacher. That is part of learning, not a problem.

Section 1: Patterns, equations and algebra thinking

Standards practised: 4.OA.A.1, 4.OA.A.2, 4.OA.A.3, 4.OA.B.4, 4.OA.C.5

A worked example

A ribbon is 3 times as long as a string. The string is 14 cm. How long is the ribbon? Write the equation.

1. 3 times as long means multiply: 3 × 14.
2. 3 × 10 = 30 and 3 × 4 = 12, so 30 + 12 = 42.
3. Equation: r = 3 × 14 = 42.

42 cm

Questions

1. Write as a multiplication: 24 is 4 times as many as 6.

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2. Maryam has 9 stickers. Noor has 5 times as many. How many stickers does Noor have?

   A. 14

   B. 4

   C. 45

   D. 54

3. A shirt costs AED 36 and a bag costs 3 times as much. How much do they cost together?

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4. 47 students go on a trip to see falcons and each bus holds 12 students. How many buses are needed?

   A. 4

   B. 3

   C. 11

   D. 35

5. List all the factor pairs of 24.

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6. Which of these numbers is prime?

   A. 27

   B. 29

   C. 21

   D. 33

7. Hind packs date cookies into boxes of 9. She has 36 cookies. Can she fill the boxes exactly with none left over? Explain.

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8. Continue the pattern: 5, 11, 17, 23, __, __. What is the rule?

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9. A pattern starts at 3 and the rule is multiply by 2. What are the first four numbers?

   A. 3, 5, 7, 9

   B. 3, 6, 9, 12

   C. 3, 6, 12, 18

   D. 3, 6, 12, 24

10. Ali scores 8 more points than Sami, who scores 3 times as many as Hind. Hind scores 6. How many does Ali score?

Section 2: Shapes, measuring and space

Standards practised: 4.MD.A.1, 4.MD.A.2, 4.MD.A.3, 4.MD.C.5, 4.MD.C.6, 4.MD.C.7, 4.G.A.1, 4.G.A.2, 4.G.A.3

A worked example

A rectangular football pitch for children is 40 m long and 25 m wide. Find the perimeter and the area.

1. Perimeter = 2 x (40 + 25) = 2 × 65 = 130 m.
2. Area = 40 × 25 = 1,000 square metres.

Perimeter 130 m; area 1,000 square metres

Questions

1. A rectangular mat in a model of a souq stall has an area of 56 square cm and a length of 8 cm. What is its width?

   A. 48 cm

   B. 448 cm

   C. 7 cm

   D. 64 cm

2. Convert: 3 km = ___ m

3. The film lasts 2 hours 30 minutes. How many minutes is that?

4. A rope is 2 m 40 cm and another is 1 m 85 cm. How much longer is the first, in cm?

5. Khalid and his cousins walk once all the way round a rectangular park that is 120 m long and 80 m wide. How far do they walk?

6. The minute hand of a clock moves from 12 to 3. Through what angle does it turn?
   A. 180 degrees
   B. 90 degrees
   C. 45 degrees
   D. 30 degrees

7. Two angles sit side by side on a straight line. One is 120 degrees. What is the other?
   A. 240 degrees
   B. 90 degrees
   C. 180 degrees
   D. 60 degrees

8. How many lines of symmetry does a rectangle that is not a square have?

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9. A triangle has angles of 90, 45 and 45 degrees. What kind of triangle is it?
   A. Right triangle
   B. Acute triangle
   C. Obtuse triangle
   D. Equilateral triangle

10. The two tracks of the metro line run side by side along the road and never meet. Are they parallel or perpendicular?

Section 3: Data, graphs and chance

Standards practised: 4.MD.B.4

A worked example

A line plot shows rope lengths in quarters of a metre: 1/4 m has 3 dots, 2/4 m has 4 dots, 3/4 m has 1 dot. What is the total length of the 2/4 m ropes?

1. There are 4 dots at 2/4.
2. 4 × 2/4 = 8/4.
3. 8/4 is the same as 2 metres.

2 metres

Questions

1. A line plot shows rope lengths: 3 dots at 1/4 m, 4 dots at 2/4 m and 1 dot at 3/4 m. How many ropes are there in all?

   A. 4

   B. 8

   C. 3

   D. 6

2. In the same line plot (3 dots at 1/4 m, 4 dots at 2/4 m, 1 dot at 3/4 m), which length is the most common?

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3. In that line plot, how much longer is the longest rope than the shortest?

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4. Hamdan makes a line plot of ropes in quarters of a metre: 2 dots at 1/4 m and 3 dots at 2/4 m. What is the total length of all these ropes?

   A. 5/4 m

   B. 5/8 m

   C. 2 m

   D. 8/8 m

5. A line plot of rainfall in centimetres, measured to the nearest half, has 3 dots at 1/2 and 2 dots at 1. What is the total?

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6. A line plot shows ribbons: 2 dots at 1/4 m, 3 dots at 1/2 m and 1 dot at 3/4 m. How many ribbons are longer than 1/4 m?

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7. A line plot shows cups of karak: 3 dots at 1/8 litre and 2 dots at 3/8 litre. How much karak is there in all?

   A. 5/8 litre

   B. 9/16 litre

   C. 1 litre

   D. 1 and 1/8 litres

8. At a family gathering 12 people chose karak, 9 chose juice and 6 chose water. How many more people chose karak than water?

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9. In a survey of 27 people, 12 chose karak, 9 chose juice and 6 chose water. What fraction chose juice, in simplest form?

   A. 1/3

   B. 3/9

   C. 9/18

   D. 1/9

10. Seven plants measure 1/2, 1/2, 3/4, 3/4, 3/4, 1, and 1 and 1/4 of a metre. How many measure 3/4 m?

Section 4: Numbers and calculating

Standards practised: 4.NBT.A.1, 4.NBT.A.2, 4.NBT.A.3, 4.NBT.B.4, 4.NBT.B.5, 4.NBT.B.6, 4.NF.A.1, 4.NF.A.2, 4.NF.B.3, 4.NF.B.4, 4.NF.C.5, 4.NF.C.6, 4.NF.C.7

A worked example

What is 3/8 + 2/8? What is 3/4 as a fraction with 8 as the denominator?

1. The denominators match, so add the numerators: 3 + 2 = 5, giving 5/8.
2. For 3/4: multiply the top and bottom by 2: 6/8.

5/8 and 6/8

Questions

1. Write in words: 45,208

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2. A solar park made 36,482 units of electricity in a month. Rounded to the nearest thousand, how many units is that?

   A. 36,000
   B. 36,500
   C. 37,000
   D. 30,000

3. The souq sold 4,325 dates on Friday and 2,678 on Saturday. How many dates is that in all?

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4. A date farm has 6,000 dates and sells 2,457. How many dates are left?

   A. 4,457
   B. 3,643
   C. 3,553
   D. 3,543

5. Multiply: 23 × 14

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6. Four cousins share AED 852 equally. How many dirhams does each cousin get?

   A. 203

   B. 231

   C. 213

   D. 2,013

7. Which fraction is equal to 2/3?

   A. 4/6

   B. 2/6

   C. 3/4

   D. 4/9

8. Noura used 4/10 of a bag of flour on Monday and 5/10 of the bag on Tuesday. What fraction of the bag did she use?

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9. Omar drinks 2/5 of a litre of milk each day. How much milk is that in 3 days?

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10. Which decimal is equal to 3/10 + 4/100?

    A. 0.7

    B. 0.43

    C. 0.034

    D. 0.34

11. Layla has four ribbons that are 0.38 m, 0.4 m, 0.309 m and 0.299 m long. Which ribbon is the longest?

    A. 0.38 m

    B. 0.4 m

    C. 0.309 m

    D. 0.299 m

12. In the number 7,700, how many times as much is the first 7 worth as the second 7? Explain.

Answer guide for parents

Section 1: Patterns, equations and algebra thinking

1. 4 × 6 = 24, because times as many means multiply 4 by 6.
2. C. 45. because 5 times as many means 5 × 9 = 45. If your child chose A (14) they added, and if they chose B (4) they subtracted.
3. 3 × 36 = 108 is the bag, then 108 + 36 = 144, so AED 144 together.
4. A. 4. 47 / 12 = 3 with 11 left over, and those 11 students still need a bus. If your child chose B (3), they ignored the remainder.
5. 1 × 24, 2 × 12, 3 × 8 and 4 × 6 all make 24, and no other whole numbers pair up to 24.
6. B. 29. Its only factors are 1 and 29. If your child chose A (27), they missed that 3 × 9 = 27.
7. 9 × 4 = 36, so 36 is a multiple of 9 and she fills 4 boxes with none left.
8. Each number is 6 more: 23 + 6 = 29 and 29 + 6 = 35. The rule is add 6.
9. D. 3, 6, 12, 24. Multiply each number by 2: 3 × 2 = 6, 6 × 2 = 12, 12 × 2 = 24. If your child chose B (3, 6, 9, 12), they added 3 each time instead.
10. Sami scores 3 × 6 = 18. Ali scores 18 + 8 = 26.

Section 2: Shapes, measuring and space

1. C. 7 cm. Width = area / length = 56 / 8 = 7. If your child chose A (48), they subtracted 8 from 56 instead of dividing.
2. 1 km is 1,000 m, so 3 × 1,000 = 3,000 m.
3. 2 hours is 2 × 60 = 120 minutes, and 120 + 30 = 150 minutes.
4. 2 m 40 cm is 240 cm and 1 m 85 cm is 185 cm, so 240 - 185 = 55 cm.
5. Once round is the perimeter: 2 x (120 + 80) = 2 × 200 = 400 m.
6. B. 90 degrees. From 12 to 3 is a quarter of the clock, and a quarter turn is 90 degrees. If your child chose A (180), they used a half turn.
7. D. 60 degrees. A straight line is 180 degrees, so 180 - 120 = 60. If your child chose C (180), they forgot to take away the 120.
8. A fold down the middle matches and a fold across the middle matches, so there are 2. A diagonal fold does not match.
9. A. Right triangle. It has one angle of exactly 90 degrees. If your child chose B (acute), they looked at the two 45 degree angles and missed the 90.
10. Parallel lines run side by side and never meet. Perpendicular lines meet at a right angle.

Section 3: Data, graphs and chance

1. B. 8. Each dot is one rope, so 3 + 4 + 1 = 8. If your child chose A (4), they counted only the dots at 2/4 m.
2. 2/4 m has 4 dots, more than the 3 dots at 1/4 m or the 1 dot at 3/4 m, so it is the most common.
3. The longest is 3/4 m and the shortest is 1/4 m, so 3/4 - 1/4 = 2/4 m.
4. C. 2 m. 2 × 1/4 = 2/4 and 3 × 2/4 = 6/4, so 2/4 + 6/4 = 8/4 = 2 m. If your child chose A (5/4), they counted every rope as 1/4 m.
5. 3 dots at 1/2 make 3 × 1/2 = 1 and 1/2. 2 dots at 1 make 2, and 1 and 1/2 + 2 = 3 and 1/2 cm.
6. Skip the 2 dots at 1/4 m. The 3 dots at 1/2 m and 1 dot at 3/4 m are longer, and 3 + 1 = 4.

7. D. 1 and 1/8 litres. 3 × 1/8 = 3/8 and 2 × 3/8 = 6/8, so 3/8 + 6/8 = 9/8 = 1 and 1/8. If your child chose A (5/8), they counted every cup as 1/8 litre.
8. 12 - 6 = 6, so 6 more people chose karak than water.
9. A. 1/3. 9 of the 27 people chose juice, and 9/27 simplifies to 1/3. If your child chose C (9/18), they left the juice drinkers out of the total.
10. Count the 3/4 values in the list: there are 3 of them, so 3 plants.

Section 4: Numbers and calculating

1. 45 thousand and 208 is said forty-five thousand, two hundred eight.
2. A. 36,000. The hundreds digit is 4, which is less than 5, so round down. If your child chose C (37,000), they rounded up.
3. Add each place and carry when needed: 4,325 + 2,678 = 7,003.
4. D. 3,543. 6,000 - 2,457 = 3,543, regrouping across the zeros. If your child chose A (4,457), they took away only the 2,000 and copied the 457.
5. 23 × 14 = 23 × 10 + 23 × 4 = 230 + 92 = 322.
6. C. 213. 852 / 4 = 213, since 4 × 213 = 852. If your child chose A (203), they put 0 in the tens place, but 5 tens / 4 gives 1 ten with 1 left over.
7. A. 4/6. Multiplying the top and bottom by 2 gives 4/6, the same size as 2/3. If your child chose B (2/6), they changed only the bottom.
8. The parts are tenths, so add only the tops: 4/10 + 5/10 = 9/10.
9. Multiply 3 days by 2/5: 3 × 2/5 = 6/5, which is 1 and 1/5 litres.
10. D. 0.34. 3/10 is 30/100, and 30/100 + 4/100 = 34/100 = 0.34. If your child chose A (0.7), they added 3 + 4 and ignored the place values.
11. B. 0.4 m. 0.4 is 0.40, or 40 hundredths, which beats 38 hundredths. If your child chose A (0.38), they thought 38 is bigger than 4.
12. The first 7 is 7,000 and the second is 700, and 7,000 / 700 = 10, so it is worth 10 times as much.